

Mohammad Bahar

516-439-3787 | MohammadBahar.ds@gmail.com | [LinkedIn](#) | [GitHub](#) |

Education

New York City College of Technology | GPA: 3.5/4.0

Expected Graduation: May 2027

Bachelor of Science in Data Science

- **Relevant Coursework:** Data Mining, Database Systems Fundamentals, Information & Data Management I–II, Probability & Statistics
- **Dean's List:** Fall 2023, Spring 2024, Spring 2026

Relevant Experience

Data Analyst Intern

Jun 2026 – Present

Metropolitan Transportation Authority (MTA)

- Cut the time to update station amenities and accessibility tracking from a few weeks to a few days, by replacing years-old manual Excel tracking with a **Power BI** dashboard covering **124** stations across **7** Metro-North lines, built with **Power Query** and **DAX**.
- Gave the stations department (about **1,000** people) and the station chief executive one dashboard for meetings and public open houses, by turning the tracking data into a single **Power BI** report.
- Replaced **30+** separate Excel workbooks with one monthly West-of-Hudson ridership report, by consolidating the data in **Excel** and correcting date errors.
- Corrected **25** errors and kept **72+** operational metrics (**2,000+** records) accurate for the MNR internal website hub, by maintaining them in a **Microsoft Access** database and writing training documentation in **Word** and **PowerPoint**.

Break Through Tech - AI/ML Fellow

May 2026 – Present

Cornell (Cornell Tech)

- Conducted exploratory data analysis and benchmarked **4 vision-document embedding models** (ColPali, SigLIP, Jina v4, Qwen) on **1,000 scanned legal documents** across 36 document-type classes, building a resumable embedding-caching pipeline in **Python**.
- Evaluated embedding quality with **three clustering algorithms** (K-Means, Agglomerative, Gaussian Mixture Models) in **scikit-learn**, using UMAP visualization and cross-validated supervised checks to narrow the model shortlist toward the top performers.

Database Engineer Intern

Jun 2025 – Aug 2025

The Difference App LLC

- Cut **10+ hours** of manual weekly adjustments and made table population **25%+** more efficient, by merging **5** separate queries (mine plus **4** teammates') into one **MySQL** query that also fixed data conflicts, and management recognized the improvement.
- Helped build and maintain the Daily Result table that powers analytics for about **10,000** users, by owning the calculation and population of the Activity Burn and Activity Calories fields in **MySQL** and contributing to metrics like **total_calories_in** and **body_fat_percentage**.
- Kept the calculated data **100%** accurate, by checking the query results against the original formulas and confirming they matched.

Projects

AI Resume-Job Matching | *Python, PyTorch, Transformers, Streamlit, Groq (Llama 3.3), Pytest*

[Website](#)

- Built a full-stack **NLP** app that scores resume-to-job fit via a four-component hybrid model (keyword matching with required/preferred weighting, semantic similarity from **768-dim sentence-transformer embeddings**, experience, and education), with a **cross-domain boost** for career-transition candidates whose keyword overlap is low but semantic fit is strong.
- Integrated the **Groq LLM API (Llama 3.3 70B)** for human-readable match explanations and built a resilient **PDF ingestion pipeline** (pypdf) that detects/repairs text-extraction errors, backed by live model resolution, caching, and multi-tier fallback handling
- Replaced a paid job-matching subscription (e.g., **Jobright AI**) with a **self-built, free alternative** running on Groq's free tier, validated with a **pytest** suite and an evaluation script across **25 sample resumes** and **5 job postings**.

Diabetes Prediction Model | *Python, XGBoost, scikit-learn, TPOT, imbalanced-learn (SMOTE)*

[GitHub](#)

- Engineered a diabetes prediction pipeline across **100,000** patient records, by benchmarking **6** classification models in **scikit-learn** and correcting an **8.6%** class imbalance with **SMOTE**.
- Selected **XGBoost** as the best-performing model, reaching **97%** accuracy and **93%** precision on a held-out test set of **~25,000** patients.
- Discovered a stronger candidate model (**97.6%** cross-validated accuracy) by using **TPOT** for automated pipeline search, which converged on a tuned **Random Forest**.

Technical Skills

Languages/Databases: Python, SQL, R, Java, JavaScript, Snowflake, MySQL, PostgreSQL, MongoDB, Pinecone

Machine Learning / AI: TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, Hugging Face Datasets, RAG, Embeddings, OpenAI API, Claude Code, Groq API, pandas, numpy, Matplotlib, Altair

Developer Tools: Git, GitHub, Bash/Terminal, Streamlit, FastAPI, pytest, Jupyter, Colab, Kaggle, Microsoft Office Suite, Power BI

Certifications: Microsoft Certified: Azure AI Fundamentals (AI-900), Machine Learning Foundations (Cornell University), SQL for Data Science (UC Davis)

Extracurricular Activities

Treasurer, Muslim Student Association: Manage club finances and event budgets; submit funding requests to Student Life & Development (SLD)

Advisor, Data Science Club: Advise on club programming and help plan events and workshops

Advisor, Google Developer Club: Help organize hackathons and club events

Fellow, CUNY Tech Prep: Data Science track, focused on machine learning and AI